Write the following intervals in symbolic form.

(i) $\{x : x \in \mathbb{R}, -2 \leq x \leq 0\}$ (ii) $\{x : x \in \mathbb{R}, 0 < x < 8\}$

(iii) $\{x : x \in \mathbb{R}, -8 < x \leq -2\}$ (iv) $\{x : x \in \mathbb{R}, -5 \leq x \leq 9\}$.

Neighbourhood

Neighbourhood of a point 'a' is any open interval containing 'a'. In particular, if ϵ is a positive number, usually very small, then the ϵ-neighbourhood of 'a' is the open interval $(a - \epsilon, a + \epsilon)$. The set $(a - \epsilon, a + \epsilon) - \{a\}$ is called ***deleted neighbourhood*** of 'a' and it is denoted as $0 < |x - a| < \epsilon$ (See Figure 1.3).



Figure 1.3

1.5 Relations

We approach the concept of relations in different aspects using real life sense, Cryptography and Geometry through Cartesian product of sets.

In our day to day life very often we come across questions like, "*How is he related to you?*". Some probable answers are,

(i) He is my father.
(ii) He is my teacher.
(iii) He is not related to me.

From this we see that the word relation connects a person with another person. Extending this idea, in mathematics we consider relations as one which connects mathematical objects. Examples,

(i) A number m is related to a number n if m divides n in $\mathbb{N}$.
(ii) A real number x is related to a real number y if $x \leq y$.
(iii) A point p is related to a line L if p lies on L.
(iv) A student X is related to a school S if X is a student of S.

Illustration 1.1 (Cryptography) For centuries, people have used ciphers or codes, to keep confidential information secure. Effective ciphers are essential to the military, to financial institutions and to computer programmers. The study of the techniques used in creating coding and decoding these ciphers is called cryptography.

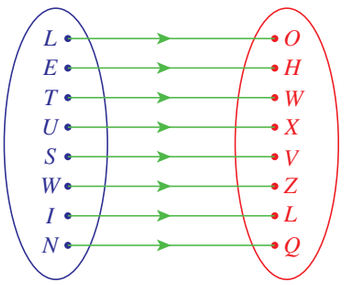


Figure 1.4

One of the earliest methods of coding a message was a simple substitution. For example, each letter in a message might be replaced by the letter that appears three places later in the alphabet.